

More on Continuous Distributions.

Example. Exponential

We say that a r.v. Y is exponential w/ parameter τ if its pdf is

$$f_Y(y) = \frac{1}{\tau} e^{-\frac{y}{\tau}} \cdot \mathbb{1}_{[0, \infty)}(y)$$

τ ... scale parameter

$$Y \sim E(\tau)$$

$$t, s > 0$$

$$\mathbb{P}[Y > t+s \mid Y > t] = ?$$

In general:

$$\mathbb{P}[Y > u] = \int_u^{+\infty} f_Y(y) dy$$

$$= \int_u^{+\infty} \frac{1}{\tau} e^{-\frac{y}{\tau}} dy = \frac{1}{\tau} \int_u^{+\infty} e^{-\frac{y}{\tau}} dy$$

$$x = -\frac{y}{\tau}$$

$$dx = -\frac{1}{\tau} dy \Rightarrow dy = -\tau dx$$

$$= \frac{1}{\tau} \int_{-\frac{u}{\tau}}^{-\infty} e^x (-\tau) dx =$$

$$= \int_{-\infty}^{-\frac{u}{\tau}} e^x dx = e^x \Big|_{-\infty}^{-\frac{u}{\tau}} = e^{-\frac{u}{\tau}}$$

$$\begin{aligned} \mathbb{P}[Y > t+s \mid Y > t] &= \frac{\mathbb{P}[Y > t+s, Y > t]}{\mathbb{P}[Y > t]} = \frac{\mathbb{P}[Y > t+s]}{\mathbb{P}[Y > t]} = \frac{e^{-\frac{t+s}{\tau}}}{e^{-\frac{t}{\tau}}} \\ &= e^{-\frac{s}{\tau}} = \mathbb{P}[Y > s] \end{aligned}$$

MEMORY LESS PROPERTY

The Normal Distribution.

$Y \sim N(\mu, \sigma^2)$ where $\mu \in \mathbb{R}$ and $\sigma > 0$
is said to be normally distributed
w/ mean μ
and standard deviation σ

if

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(y-\mu)^2}{2\sigma^2}} \text{ for all } y \in \mathbb{R}$$

If $\mu=0$ and $\sigma=1$, we say that Y is standard normal
Its pdf is

$$\varphi(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} \text{ for } y \in \mathbb{R}$$

Q: Let $Y \sim N(\mu, \sigma^2)$

$$\frac{Y-\mu}{\sigma} \sim N(0,1)$$

Q: Let $Z \sim N(0,1)$

Let α and β be two real constants

$$\alpha Z + \beta \sim \text{Normal}(\beta, \alpha^2)$$

Expectation [revisited].

In the discrete case:

$$\mathbb{E}[Y] := \sum_{y \in S_Y} y \cdot p_Y(y) \text{ if it exists}$$

Def'n. Let Y be a continuous r.v. w/ pdf f_Y .
We define the expectation of Y as

$$\mathbb{E}[Y] := \int_{-\infty}^{\infty} y \cdot f_Y(y) dy \text{ if the integral exists}$$

Task: Cauchy Dist'n.

Example. $Y \sim E(\tau)$

$$E[Y] = ?$$

$$\begin{aligned} \rightarrow: E[Y] &= \int_{-\infty}^{\infty} y f_Y(y) dy \\ &= \int_0^{\infty} y \frac{1}{\tau} e^{-\frac{y}{\tau}} dy \end{aligned}$$

$$\left[\begin{array}{ll} u = y & du = dy \\ dv = \frac{1}{\tau} e^{-\frac{y}{\tau}} dy & v = e^{-\frac{y}{\tau}} \end{array} \right]$$

$$= - \underline{y e^{-\frac{y}{\tau}}} \Big|_0^{\infty} - \int_0^{\infty} (-e^{-\frac{y}{\tau}}) dy = \tau$$

